Eric Solak

Toronto, ON · +1 (226) 388-4341 · solakeric28@gmail.com · ericsolak.me · linkedin.com/in/ericsolak · github.com/eric-solak

EDUCATION

McMaster University | Hamilton, ON **2022 - May 2026**
Bachelor of Engineering, Software Engineering
cGPA: 3.70/4

EXPERIENCE

Alloy | Toronto, ON **Jun 2026 - Present**
Founder

- Grew Alloy into a solo consultancy whose automations save client businesses **100+ hours of manual work per month**, handling all client-facing and technical work single-handedly.
- Designed and built production systems end to end: custom websites in React and TypeScript deployed on Cloudflare Workers, and AI-driven automation pipelines tailored to each business's existing systems.
- Built **fully and semi-autonomous AI workflows** automating invoicing, document summarization, sales tracking, and commission calculations for small and mid-sized businesses.
- Architected integration layers connecting client automations to **accounting** (QuickBooks), **CRM and commerce** (Salesforce, Shopify, Stripe), and **productivity tools** (Microsoft 365, Google Workspace), handling authentication, data mapping, and error recovery across disparate systems.
- Delivered automation and website projects for businesses across Canada, owning requirements gathering, scoping, build, and handoff within a **1-3 week window per project**.
- Validated automations against live client data before handoff, iterating on edge cases found through real usage.
- Sustained an **80%+ client retention and referral rate** through direct, hands-on relationship management.

CATTLElytics Inc. | Hamilton, ON **Sep 2025 - Apr 2026**
Industry Capstone Project

- **Led AI development** for Pocket AI, an AI-powered hands-free app for workers who can't stop to type: it listens to natural speech to create and manage tasks in real time. Delivered as a complete product to CATTLElytics, an industry-leading global agri-tech company.
- **Directed a team of 5 engineers** through the full product development lifecycle in **8 months**, from requirements gathering and architecture design to production handoff, delivering a fully functional application now being integrated into CATTLElytics' platform.
- Integrated OpenAI Whisper for real-time audio transcription, validated through usability testing that achieved **98% system voice recognition accuracy** across 200 test commands under simulated and true barn noise.
- Built the Flask backend powering voice-to-task workflows, with REST APIs for authentication and task lifecycle management and an offline-first architecture for reliable performance in low-connectivity farm environments.
- Designed and built mobile UI components in React Native (Expo), including task views, audio capture flows, and speech confirmation feedback; usability testing yielded a **mean satisfaction score of 4.4/5**.

PROJECTS

musicGPT · 45M-Parameter LLM + RAG from Scratch | Python, PyTorch **Jul - Oct 2026**

- Built a **45M-parameter decoder-only transformer GPT entirely from scratch**, hand-implementing RoPE, causal self-attention, LayerNorm, and the training loop.
- Trained on a 0.92B-token pipeline with a hand-rolled markup stripper and 3x upweighting of music-domain content; **reduced val loss from 10.9 (uniform baseline) to 3.26** with no overfitting across 14,000 steps.
- Fine-tuned the model into a **grounded music Q&A system** with a hand-written BM25 search over 14,278 Wikipedia articles: on 400 held-out questions, **99% of the answers it gave were correct** (337 of 340).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, SQL, C
Frameworks & Libraries: PyTorch, FastAPI, Flask, React Native, NumPy, Pandas, scikit-learn
Tools: Docker, Render, Claude Code, GitHub Actions, JUnit, SQLite

INTERESTS

Self-taught musician (piano & guitar, 10 years) · chess · weightlifting · running